

MODULE 05

Integrations & Crons

Connecting to the

Real World

1Password, Google Calendar, YouTube, RapidAPI and
automated crons

DURATION

25 min

FORMAT

Live coding + demo

KIT

3 files

MODULE 05 · INTEGRATIONS & CRONS

Build Your AI COO

1Password: Secure Credentials

Before ANY integration, configure the 1Password CLI. NEVER hardcode credentials.

Installing 1Password CLI

How to Use in Code

Instead of hardcoding API keys:

CAUTION: systemd Override

If the gateway runs via systemd, the .service file can override the .env .

Update BOTH when changing credentials.

Install via official script

```
curl -sS https://downloads.1password.com/linux/keys/1password.asc | \
sudo gpg --dearmor --output /usr/share/keyrings/1password-archive-keyring.gpg
echo "deb [arch=amd64] https://downloads.1password.com/linux/debian/amd64 stable m
sudo tee /etc/apt/sources.list.d/1password.list
sudo apt update && sudo apt install 1password-cli -y
```

Login

op signin

BASH

// BAD!

```
const apiKey = "sk-ant-1234567890abcdef";
```

// · GOOD!

```
const apiKey = process.env.ANTHROPIC_API_KEY;
```

JAVASCRIPT

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Build Your AI COO

Google Calendar & Drive (GOG CLI)

GOG CLI allows direct access to Google APIs. Ideal for Calendar, Drive, Gmail.

YouTube: Data API + OAuth

To access YouTube (upload, analytics, comments), you need OAuth2.

Install GOG CLI (via skill or manual)

openclaw skills install gog

Authenticate

gog auth login

Test

gog calendar list

BASH

Create Project in Google Cloud

Access console.cloud.google.com · Create project

1

Enable YouTube Data API v3

APIs & Services · Enable APIs · YouTube Data API v3

2

Create OAuth Credentials

Credentials · Create · OAuth client ID · Desktop app

3

Download JSON and Authenticate

Save `client_secret.json` on the server · run OAuth flow

4

..Daily Quota

YouTube Data API has a quota of 10,000 units/day. 1 upload = ~1,600 units.

Plan your usage.

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MODULE 05 · INTEGRATIONS & CRONS

Build Your AI COO

RapidAPI: Universal Proxy

Cloud IPs (VPS) are blocked by YouTube, Instagram, X. RapidAPI solves this by proxying the requests.

Useful APIs on RapidAPI

API

WHAT IT DOES

Instagram Statistics

Get metrics from posts/profiles without an official API

X/Twitter API45

Access tweets, profiles, trends

YouTube Transcripts

Extract video captions

Telegram As An Operational Hub

Use Telegram with group + topics as your "Mission Control".

Why Telegram > WhatsApp?

Telegram: Multiple simultaneous sessions (topics = war rooms)

WhatsApp: Single session (everything mixed)

Telegram: Native bots, without API limitations

WhatsApp: Needs workarounds (Baileys, etc)

·Topic Structure

Create separate topics: "Daily Briefing", "Urgent", "Projects", "Metrics". Each cron delivers to the right topic.

Example: search for video transcript

```
curl -X GET \
```

```
"https://youtube-transcripts.p.rapidapi.com/transcript?video_id=dQw4w9WgXcQ" \
```

```
-H "X-RapidAPI-Key: YOUR_KEY"
```

BASH

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MODULE 05 · INTEGRATIONS & CRONS

Build Your AI COO

Crons: Automate Recurring Tasks

Crons are tasks that run at specific times. OpenClaw has an integrated cron system.

The Bug EVERYONE Finds

systemEvent + main = Does NOT Work As Expected

If you configure cron with systemEvent in the main session, it triggers but DOES NOT execute (durationMs ~0ms).

CORRECT Solution:

Why?

sessionTarget: isolated · Creates its own session for the cron

agentTurn · Triggers the agent's turn (it EXECUTES)

announce · Sends the result to the configured channel

Other Common Problems

```
{
  "schedule": "0 9 * * *",
  "sessionTarget": "isolated",
  "agentTurn": {
    "announce": true,
    "message": "Execute o daily briefing e entregue no tópico Daily Briefing."
  }
}
```

JSON

Schedule Collision

Crons at the same time cause problems. Space 15-30min between each.

1

config.patch Restarts Gateway

When changing config via config.patch , the gateway restarts and kills crons in execution.

Do it at times without active crons.

2

Reminders Don't Notify

3

systemEvent does NOT send notification. Use agentTurn + message send if you need to notify.

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MODULE 05 · INTEGRATIONS & CRONS

Build Your AI COO

Cron Example: Daily Briefing

Split Model: Real Economy

TYPE OF

CRON

RECOMMENDED

MODEL

WHY

Execution

Sonnet

Sync, watchdog, reminders · simple tasks

Analysis

Opus

Strategic reports, complex decisions

Heartbeats

Haiku

Periodic checks · 90% economy

·Result

With this strategy, you reduce costs from ~\$2-3/day to ~\$0.10/day.

Useful Commands

```
{
  "name": "daily-briefing",
  "schedule": "0 9 * * *",
  "sessionTarget": "isolated",
  "agentTurn": {
    "announce": true,
    "targetChannel": "telegram",
    "targetId": "YOUR_GROUP_ID",
    "targetTopic": "Daily Briefing",
    "message": "Execute the daily briefing. Include: events of the day, pending tasks"
  },
  "model": "claude-sonnet-4"
}
```

JSON

List active crons

openclaw crons list

See next executions

openclaw crons upcoming

Test cron manually

openclaw crons run daily-briefing

BASH

Pause cron

openclaw crons pause daily-briefing

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MODULE 05 · INTEGRATIONS & CRONS

Build Your AI COO

Module 5 Checkpoint

1Password CLI configured (zero hardcoded)

At least 1 integration working (Calendar/Drive/YouTube)

2 crons configured (isolated + agentTurn)

Telegram with topics as operational hub

Split model configured (Haiku/Sonnet/Opus)

· Prompt for the Agent

Paste this prompt into your OpenClaw chat after watching Module 5:

I just watched Module 5 of the course on integrations and crons. Read the PRD of integrations and guide me. What I need you to do: 1. Analyze my profile · based on what you already know about me (USER.md), recommend the 3-5 most useful integrations for start 2. Guide me in installing each integration, one at a time: - Explain what it does and why it's useful - Help me securely configure the API key - Test if it worked 3. Configure at least 2 crons: - A reminder/schedule (for me to see it working) - An automatic report (daily briefing or metrics) - ALWAYS use: sessionTarget: isolated + agentTurn + announce (this is critical to working!) 4. Explain the common errors: - Why systemEvent in the main session doesn't work as expected - Why crons at the same time cause problems - How config.patch affects running crons Rules: - Start with simple integrations (calendar, weather) before complex ones - If there is a cloud IP blocked error, explain the solution to me (RapidAPI as proxy) - ALL credentials go to 1Password · zero hardcoded in the files - At the end, show me the active crons and when they will run Shall we connect to the real world?

·

NEXT MODULE

Module 6 · Skills: Instant Superpowers

Module 05 · Integrações & Crons

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